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Regulating Conflicts of Interest in Medicine Through Public Disclosure: Evidence from a Physician Payments Sunshine Law

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Abstract. Hospital and healthcare administrators name high prescription drug costs as one of their largest problems. A significant body of research demonstrates that meals and honoraria from pharmaceutical firms to physicians leads to higher prescribing of expensive, brand name drugs, despite little difference in efficacy. Some administrators and scholars have advocated for mandatory disclosure of these payments in order to reduce this conflict of interest, but many practitioners believe disclosure has little effect on prescribing, and the empirical evidence is mixed. This paper uses a quasi-experiment of a 2009 payment disclosure policy in Massachusetts to estimate the causal impact of public disclosure on prescribing. The comprehensive data set includes all retail prescriptions for 262 drugs in nine drug classes written by 5,730 physicians in five states over 48 months. We show a significant postdisclosure reduction in brand name drug prescriptions by Massachusetts physicians, relative to control physicians in other states. These effects are driven by heavy prescribers of brand name drugs in the prepolicy period, particularly for drugs with large prepolicy sales forces. Effects are also detected before the first data were released, implying that the effects are not because patients or administrators responded to the disclosed payments. Instead, some physicians may have changed their payments and prescriptions behavior to avoid appearing biased. Taken in tandem with the many studies showing that pharmaceutical industry payments influence prescribing, this study suggests a strong role for mandatory public disclosure in reducing conflicts of interest in medicine and costly prescribing of brand name drugs.

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Keywords: conflicts of interest • disclosure • social image • pharmaceutical marketing • detailing

1. Introduction

Hospitals, health center administrators, insurance companies, and regulators have long struggled with the tradeoffs inherent in pharmaceutical marketing to physicians. Marketing to doctors and other forms of industry-physician relationships can yield mutually beneficial information flow and research collaborations (e.g., Stossel 2008, Chintagunta et al. 2012, Huang et al. 2019), but they also introduce conflicts of interest when industry offers incentives, such as sponsored meals or consulting arrangements, to facilitate interactions (Fugh-Berman and Ahari 2007). Annual industry spending on these marketing incentives have typically exceeded industry spending on research and development of new drugs (Gagnon and Lexchin 2008), and there is now “overwhelming” evidence (Brennan et al. 2006), including meta-analyses (Lexchin 1993, Wazana 2000, Spurling et al. 2010) and quasi-experimental evidence (Carey et al. 2015,

Larkin et al. 2017, Engelberg et al. 2014), that industry-sponsored meals and consulting payments can increase prescriptions of branded, expensive drugs, perhaps by as much as 10% (Larkin et al. 2017).

This increased prescribing of marketed drugs can greatly affect healthcare costs. New branded drugs are the most expensive, averaging a retail cost of \$5,800 per drug per year when taken on a chronic basis (Schondelmeyer and Purvis 2016), which is 2,000% higher than older drugs (Klingler 2017). Despite this, research demonstrates that many newer, expensive drugs are not better on average than other alternatives, leading many to be labeled as “me too” drugs (Régner 2013, Sacks et al. 2018). Nevertheless, these drugs are often the most heavily marketed and most prescribed (Kamal et al. 2019). For instance, Lipitor and Nexium are often classified as me too drugs, yet they had two of the largest sales forces in the United States in the late 2000s and were the two

best-selling drugs by revenue in 2009 (Spector 2005, Bartholow 2010, Greene 2010). Prescription drugs now account for an estimated 15%–17% of total health-care expenditures, with 72% of that attributed to brand name drugs (Kesselheim et al. 2016, American Academy of Actuaries 2018). Not surprisingly, hospital and other healthcare delivery executives have listed “managing costs, with a focus on drug spending” as the top overall area for improvement required by their organizations, above priorities such as “tracking and reporting clinical quality” or “engaging consumers in health care decisions” (White 2017).

Public disclosure is frequently touted as a way to reduce marketing influence on prescriptions, because it can provide transparency without necessarily disrupting productive interactions (Steinbrook 2009). After all, evidence exists that industry relationships provide faster dissemination of useful drug information (Huang et al. 2019); more strict policies, such as banning meals or consulting payments, could hinder industry’s ability to share useful information and also disrupt consulting payments for legitimate research. Many institutions, states, and the federal government have therefore implemented disclosure policies of varying strictness, with some disclosing to the government, others to the public, and some covering a wider range of payments than others (e.g., Chimonas et al. 2010). For healthcare officials concerned with marketing influence on prescription costs, further widening the scope of disclosure may be a solution.¹

However, there is limited evidence on whether disclosure has any impact in this context (Armstrong and Freiberg 2017). Advocates of disclosure often assume that transparency will allow patients to make better informed decisions (Steinbrook 2009), as predicted in rational models that treat the patient as the “principal” who relies on an expert “agent” for advice when making a decision (Crawford and Sobel 1982). However, laboratory evidence on principal-agent contracts suggest that principals often ignore disclosed conflicts of interest or otherwise insufficiently adjust their beliefs (Cain et al. 2005, Hampson et al. 2006, Loewenstein et al. 2011). Indeed, surveys show that up to 95% of patients whose physicians are subject to disclosure are unaware of whether their physician accepts payments from industry (Pham-Kanter et al. 2017). Many experts thus conclude that disclosure laws are failing (Lenzer 2016). This has led to calls for more draconian solutions, such as restricting salesperson access to physicians, which many hospitals have implemented to varying degrees (e.g., Larkin et al. 2017) despite possible negative effects on the flow of useful information (Chressanthis et al. 2012).

Nevertheless, there are behavioral reasons why disclosure can still impact outcomes. Laboratory evidence shows that if advisors do not want to appear

conflicted to others, they may abstain from incentives that create conflicts of interest, leading to subsequent changes in their advice (Sah and Loewenstein 2014). This predicts that disclosure can still curb biased actions from physicians, but only if physicians choose to reduce or abstain from conflicting incentives in order to reduce the appearance of impropriety, subsequently leading to changes in their actions, that is, their prescriptions. Alternatively, physicians may choose to continue accepting payments but to reduce prescriptions of marketed drugs as another way to avoid appearing biased.

This paper tests for the effects of the 2009 Massachusetts mandatory public disclosure policy that required pharmaceutical companies to disclose free meals and consulting payments made to physicians. This policy required that all such transactions above a minimum amount be made publicly available in an online, searchable database. We test whether disclosure has any effect on prescriptions, and if so, whether it is through influencing advisors as in Sah and Loewenstein (2014), or by influencing patients as assumed in rational models (Crawford and Sobel 1982). Using a data set of 5,730 physicians across treatment and control states, and covering prescriptions for 262 drugs and 48 months, we estimate that disclosure caused physicians to reduce prescriptions of branded drugs, thus reversing some of the documented effects of industry marketing on prescriptions.

These reductions in branded prescriptions are predominantly observed in the drugs and doctors most likely to have been affected by pharmaceutical meals and payments. First, the most affected drugs were those that had a dedicated national sales force just prior to policy implementation, indicating they were the most highly marketed drugs at the time. Second, we split doctors based on their prepolicy prescribing volume, because pharmaceutical salespeople are instructed to primarily target high prescribing doctors (Fugh-Berman and Ahari 2007). We find that only high prescribers showed any decrease in branded prescriptions. Our results therefore suggest that disclosure affected precisely those drugs and doctors that were most targeted by pharmaceutical marketing.

Our results also provide evidence that disclosure effects are not patient-driven. Effects are detectable before the first batch of data were published or when excluding physicians that had anything to disclose. This suggests that the effects are not because patients (or colleagues, or any others) were responding to a physician’s disclosed payments and pressuring the physician to change.

Instead, our results are consistent with an agent-based response. For instance, physicians may be reducing or opting out of industry meals and payments to avoid appearing biased to others. This would decrease branded prescriptions, as documented by other

research (e.g., Spurling et al. 2010, Larkin et al. 2017). Consistent with this, restaurants lobbied hard against the policy, and in 2012 a representative of the Restaurant and Business Alliance stated that the policy “is costing our state tens of millions of dollars in sales taxes, income taxes, meals taxes, and convention revenue” (Sullivan 2012). Opting-out is also consistent with laboratory experiments (Sah and Loewenstein 2014) and with correlational evidence that fewer physicians accepted payments in states that had disclosure (Chen et al. 2019). Alternatively, some physicians may continue accepting payments but still reduce prescriptions of marketed drugs to avoid appearing biased.²

These results have important implications for managing conflicts of interest in healthcare. Although some have argued that lack of patient engagement with disclosures data implies these laws are failing (Lenzer 2016), our paper suggests that disclosure can still impact prescriptions by influencing physicians. Health executives attempting to curb pharmaceutical influence on drug costs may consider widening the scope of payments covered by disclosure. They may also consider making the disclosed payments more salient to physicians (e.g., through hospital-wide communications or campaigns) to further motivate physicians to appear unconflicted. In addition, they may consider using disclosure for other conflicting incentives beyond just those from pharmaceuticals. For instance, some doctor-driven movements have advocated disclosing incentives for ordering additional tests or procedures (Wen 2013a,b; Sifferlin 2014), and our results add insight on which mechanisms could matter if disclosure is implemented in these contexts.

This paper is organized as follows. Section 2 reviews the industry background and related literature. Section 3 summarizes the overall empirical approach and data. Section 4 presents the paper’s empirical results, including comprehensive robustness checks. Section 5 concludes.

2. Background

2.1. Healthcare Costs and Physician-Industry Interactions

The pharmaceutical industry spends billions of dollars annually to market their drugs to physicians, with spending on marketing often outpacing spending on research and development (Gagnon and Lexchin 2008, Foley 2019). These marketing expenditures come primarily in the form of tens of thousands of sales representatives, who pay biweekly “detailing” visits to physician offices armed with free meals, gifts, and occasionally consulting payment arrangements (Fugh-Berman and Ahari 2007, Campbell et al. 2010, Rockoff 2012). Recent research has causally tied these detailing activities to the prescribing of expensive, marketed drugs (Carey et al. 2015, Larkin et al. 2017, Engelberg et al. 2014, Grennan et al. 2018). There is

also significant correlational evidence in the medical (Lexchin 1993, Wazana 2000, Spurling et al. 2010) and marketing literatures (Rizzo 1999, Manchanda and Chintagunta 2004, Mizik and Jacobson 2004).

However, evidence suggests that over 90% of new drugs are no more efficacious for the median patient than older drugs (Régner 2013). For instance, Nexium was the second highest prescribed branded drug in 2009 (Bartholow 2010), had the largest pre-disclosure sales force size (4,155) among drugs in this paper’s data set, and yet it is considered biochemically almost identical to the generic drug omeprazole (Spector 2005), with little evidence suggesting it is more effective (ibid). Despite these similarities, the average cost difference between newer and older drugs is over 2,000% (Klingler 2017). To an individual patient on insurance, the cost difference is a small fraction of that, but to the overall healthcare system, \$1.7 trillion has been spent in the last 10 years on the newest marketed drugs (Klingler 2017), with spending on these drugs continuing to rise the fastest among prescription drugs (Kamal et al. 2019).

As a result, the cost implications of conflicts of interest from industry marketing is potentially enormous. An estimated 15% of total healthcare spending in the United States goes to the cost of prescription drugs (up from 7% in the 1980s), with 72% of these drug costs attributed to brand name drugs despite representing only 10% of total drugs dispensed (Kesselheim et al. 2016, American Academy of Actuaries 2018). For these reasons, hospital and other healthcare executives have listed reducing costs associated with prescription drugs as the top overall area for improvement required by their organizations (White 2017).

2.2. Disclosure Policies as a Remedy

The most common policy solution has been mandatory disclosure of payments (e.g., Chimonas et al. 2010). Because physician-industry relationships may involve useful and fruitful interactions (Stossel 2008, Huang et al. 2019), disclosure is seen as a viable solution that reduces influence from payments but does not disrupt useful interactions. Many advocates of disclosure presume that patients will use the disclosed data to monitor physicians, as in rational models (Crawford and Sobel 1982); however, surveys indicate that 95% of patients do not make any use of disclosed payments data (Pham-Kanter et al. 2017).

It is therefore important to test whether disclosure can influence physician behavior instead. For instance, physicians concerned with appearing biased may opt to reduce or abstain from industry meals and payments (Sah and Loewenstein 2014), which would likely change how they prescribe (Larkin et al. 2017). Alternatively, they may also reduce prescriptions to avoid appearing biased even as they

continue accepting payments. However, disclosure may also immunize the physician in their own mind from any future conflict, increasing their bias in an effect known as “moral licensing” (Cain et al. 2005, Barbour et al. 2012). Despite concern in the medical literature over moral licensing (e.g., Barbour et al. 2012), it is unclear whether moral licensing would apply in this context, because it requires agents to view the act of disclosure as a virtue unto itself (Effron and Conway 2015). This paper aims to test for these mechanisms, while using additional drug and doctor heterogeneity to test between patient-driven versus physician-driven effects.

2.3. Differences from Existing Literature

Our paper differs from research using payments data that have been disclosed, but which does not test for the effects of disclosure. These other papers test for correlations between disclosed payments and prescriptions (e.g., Engelberg et al. 2014, Grennan et al. 2018), or they examine physician traits that correlate with having payments to disclose (e.g., Tringale et al. 2017, Brunt 2018). In contrast, our paper examines whether the policy of requiring disclosure in and of itself affected prescribing.

Our study also differs from the few studies that examined disclosure in a prescriptions context. Pham-Kanter et al. (2012) finds that disclosure policies in Maine and West Virginia had negligible to small effects on two drug classes.³ However, these policies required only disclosure to the state health department and not to the public in a searchable database (Chimonas et al. 2010). We instead test whether disclosure has different effects when it invokes image concerns via a publicly searchable database.

Parker (2017) finds that unanticipated disclosure of payments (due to court orders and whistleblower laws) had small, positive effects on branded drug prescriptions in New Jersey hospitals. However, because physicians did not know of disclosure in advance, physicians had no chance to opt out of meals and payments to avoid being disclosed. As such, some effects of disclosure (such as physicians opting out of payments) were not feasible in this context, perhaps leading to differences in effects. In addition, due to differences in context, it is also possible that moral licensing is more relevant in their setting than ours.

Guo et al. (2019) examines the same 2009 Massachusetts disclosure policy, but using physicians on the geographic borders of Massachusetts and neighboring states. They find that disclosure significantly decreased both branded and generic drug prescriptions, and they conclude that disclosure may work by making physicians more self-aware and guarded against overprescribing in general, leading to decreases in all drugs, even nonmarketed generics.

Our study finds evidence for a different mechanism from theirs, due to different results and data. We examine drug and physician heterogeneity and find that disclosure predominantly affected the drugs and doctors most likely to have been targeted by pharmaceutical salespeople. Thus, unlike Guo et al. (2019), our results provide evidence that the effects of disclosure appear related to the marketing activities and payments that had to be disclosed. Importantly, this does not rule out the possibility that some physicians also decreased prescriptions due to increased self-awareness of overprescribing, as theorized in Guo et al. (2019). However, because we do not find a decrease in generic prescriptions (as they do), our results better align with the wider literature suggesting that marketing increases branded (but not generic) prescriptions, and that policies that decrease these payments lead to drops in branded (but not generic) prescriptions (e.g., Larkin et al. 2017).⁴

Finally, relative to previous papers, ours uses larger and more detailed data on drugs, prescriptions, and marketing. For instance, our paper uses total prescriptions at the doctor-drug-month level, instead of just prescriptions from one insurance carrier. It is also the first to have any predisclosure, drug-level measures of marketing (sales force size), and to include analysis based on predisclosure doctor differences in prescribing volume. These additional details better isolate the mechanisms behind disclosure effects.

3. Empirical Approach and Data

3.1. Overall Empirical Approach

We treat the mandatory disclosure of payments to physicians by pharmaceutical companies in Massachusetts (hereafter the “sunshine law”) as a quasi-exogenous shock to public disclosure of payments. Starting July 1, 2009, the law required all payments over \$50 to be disclosed to state regulators, who published the payments annually on a state website. In practice, some pharmaceutical companies disclosed data on all payments, and the database includes many payments less than the \$50 minimum.

We compare changes to prescribing for all full-time academic medical center (AMC) affiliated Massachusetts physicians after July 2009 to changes in prescribing for full-time AMC-affiliated physicians over the same time period in four other states with significant presence of academic doctors. Due to high data costs, we were limited in the number of doctors for whom we could obtain comprehensive prescribing, and limiting to academic physicians in practice ensures the sample has physicians with similar training, background, work environment, and financial incentives. Our data run from July 2007 to June 2011, covering two full years preintervention and two full years postintervention.

3.2. Physicians and Affiliations Data

We obtained data on the affiliation of all physicians employed at AMCs in five states (Massachusetts, Pennsylvania, California, New York, and Illinois). These states were chosen because they contained the most AMCs, including many of the most prominent AMCs in the country. A full list of the AMCs included in the study is in the appendix. We obtained the list of affiliated physicians from IMS Health, a leading pharmaceutical market research firm. IMS surveys AMCs and other physician practices on a quarterly basis to obtain affiliations data. Because IMS charged on a per-physician basis, we excluded AMC physicians who had a total of 50 prescriptions or less in the study's nine drug classes during the study period. We also excluded part-time or admitting-only physicians from the study, because these physicians likely had affiliations with private medical groups and may have differed in training and work environment. After excluding physicians that IMS listed with multiple affiliations during our prepolicy period,⁵ our final list included 5,730 study physicians. Because of the prominence of academic medicine in Massachusetts, approximately half of the study doctors were in the treatment group.

3.3. Prescribing Data

We focused on nine drug classes: statins, anti-hyperglycemics, proton pump inhibitors, antihypertensives, antidepressants, anxiolytics, sleep aids, antihyperactives, and antipsychotics. These were chosen after a comprehensive set of interviews with physicians who indicated they contained a mix of heavily marketed and generic drugs. Many of these drug classes were used in previous studies on marketing or disclosure (Pham-Kanter et al. 2012, Larkin et al. 2014, Larkin et al. 2017, Guo et al. 2019). For each of these classes, all drugs within the class (according to IMS classifications) were included except for drugs

that never reached 0.1% market share in prescriptions within that drug class in a single month.

Data on prescribing at the physician-drug level were obtained from IMS Health. IMS Health has the most comprehensive database of retail prescriptions in the United States, and it is the most commonly used prescriptions data by pharmaceutical sales representatives for tracking doctor response to their payments and interactions. IMS obtains these data by purchasing it from pharmacies representing over 75% of total prescription volume. It uses that data to estimate prescriptions for the remaining pharmacies (by volume). The data do not include hospital pharmacies (unless the hospital pharmacy was outsourced to a retail pharmacy chain), and they also do not include ordered but ultimately unfilled prescriptions. The data only indicate the ultimate drug dispensed to the patient, regardless of what was written on the prescription script. Most insurance policies require mandatory interchange to generic drugs when a bioequivalent generic is available, so even if a physician writes a script using the drug's branded name, the resulting drug dispensed is the generic equivalent.⁶ The data were purchased in two batches, with the second batch including some drugs that were new to the market and dropping some that were no longer prescribed often.

Table 1 lists the top five most prescribed branded drugs and top five most prescribed generic drugs, for both prepolicy and postpolicy periods. The most prescribed branded drugs (both prepolicy and postpolicy) are Lipitor, Seroquel, Lexapro, Diovan, and Nexium, in that order. These five drugs all ranked in the top 30 by U.S. sales in 2009, with three of them placing in the top six (Bartholow 2010). Lipitor and Nexium are infamous for being me too drugs, with Lipitor being highly similar in chemical composition and method of action to simvastatin, and Nexium being highly similar to omeprazole (Spector 2005, Greene 2010).

Table 1. Most Prescribed Drugs, Top 5

	Prepolicy	Total Rx	Postpolicy	Total Rx
Branded Drugs	Lipitor	422,009	Lipitor	324,775
	Seroquel	165,287	Seroquel	154,172
	Lexapro	122,899	Lexapro	115,577
	Diovan	102,113	Diovan	113,457
	Nexium	91,720	Nexium	94,840
Generic Drugs	Lisinopril	537,423	Lisinopril	621,376
	Atenolol	345,769	Omeprazole (Rx ^a)	372,757
	Omeprazole (Rx ^a)	234,632	Atenolol	321,705
	Metformin HCl	229,714	Metformin HCl	272,755
	Metoprolol Tartrate	222,615	Citalopram HBr	269,332

Note. Aggregated across all doctors (treatment and control) in the data set.

^aPrescription strength omeprazole only (not over the counter).

3.4. Drug Marketing Data

For our data through June 2009, we purchased data on detailing and national sales force size for all the non-generic drugs in the database, as estimated by the market research firm SDI Health. These data indicated that 95% of the branded drugs with no generic alternative available were still detailed or marketed in some fashion.

We use the predisclosure sales force data as proxies for whether a drug is more heavily marketed. Although the sales force data only pertain to predisclosure periods at the national level, they serve as a reasonable proxy for differing degrees of marketing. For instance, Nexium, the second highest prescribed branded drug in 2009 (fifth-most in our data), had the largest sales force size in our data for the second quarter of 2009, at 4,155 sales reps. Because the data are national, and because we only have sales force data up through June 2009, our measure could erroneously categorize a few drugs as having a sales force postpolicy. To the extent that this occurs, this would likely bias against finding effects.

3.5. Massachusetts Payments Disclosures

Industry gifts, meals, and payments to Massachusetts physicians for July 2009–December 2011 are publicly available. This data set includes the name of each physician and the value and type of compensation they received (e.g., consulting fees or meals) in a given year. These data were merged to the IMS physician affiliation data set using first name, last name, and middle initial or middle name. Work address was also used in the rare cases (< 0.5%) where matching by name led to duplicate matches. Any mismatches would likely bias against finding effects.

Table 2 summarizes the disclosures data. In total, 424 out of the 2,897 Massachusetts physicians in our data set appear in the disclosures data between July 2009 and December 2011 for accepting meals, and 394 physicians appear for accepting consulting payments. These two payment types represent over 70% of total payments (by individual transaction) within the state's disclosures data during these years.

3.6. Data Exclusions

When analyzing the data, we exclude several categories of observations from the data. This is done to avoid confounds. (Figure A.1 in the appendix summarizes these in table form.)

First, among the disclosed payments, we dropped payments for educational grants (13.7% of total transactions), education and training (9.9%), and several small categories representing less than 3% each of total transactions. These transactions are explicitly about information flow, and the majority of spending on them is for informational materials, conference fees and the like, and not direct gifts to physicians.

Next, we drop all branded drugs where a generic version was introduced in the middle of this time period. When a generic drug becomes available, insurance companies almost always require that the generic be prescribed even if the physician writes a script for the biochemically equivalent branded drug.⁷ This causes prescriptions of the branded drug to drop drastically (and it is also why drugs are no longer marketed by salespeople once a generic becomes available). Because this represents such a sharp discontinuity in prescribing, we chose to leave drugs affected by sudden generic availability out of the

Table 2. Payments Data Summary (Massachusetts)

Sample	Accepted Meals				Accepted Consulting			
	Physician count	Average \$ ^a	Median \$ ^a	Range ^a	Physician count	Average \$ ^a	Median \$ ^a	Range ^a
All: July 2009–December 2011	424	\$335.58 (561.21)	\$145.50	\$12.00 to \$5,326.33	394	\$20,354.86 (39,277.97)	\$7,007.00	\$56.00 to \$375,248.40
By prescribing volume ^b								
Above median	235	\$385.21 (698.69)	\$145.00	\$12.00 to \$5,326.33	228	\$22,131.51 (45,099.40)	\$6,686.50	\$59.72 to \$375,248.40
Below median	189	\$273.86 (306.82)	\$146.00	\$50.00 to \$1,989.94	166	\$17,914.64 (29,427.66)	\$7,334.50	\$56.00 to \$247,456.20
By sales force ^c								
Prescribed	350	\$347.31 (596.56)	\$146.00	\$12.00 to \$5,326.33	327	\$21,051.93 (40,224.30)	\$7,036.90	\$59.72 to \$375,248.40
Never prescribed	74	\$280.06 (346.17)	\$142.26	\$51.38 to \$1,601.52	67	\$16,952.73 (34,356.32)	\$4,050.00	\$56.00 to \$247,456.20

^aThe means and medians are calculated based on only those who actually accepted payments.

^bThis splits the 2,897 Massachusetts physicians into a top and bottom half, based on total scripts prescribed during the prepolicy period.

^cThis splits Massachusetts physicians based on whether they prescribed any drug during the prepolicy period that had a national sales force size > 0.

There were 686 out of 2,987 Massachusetts physicians (24%) who never prescribed a drug, prepolicy, with a national sales force.

database. Because these branded drugs are not included, we also exclude their generic versions that entered the market during this time. The appendix reports results with all of these drugs still included; keeping them yields similar results.

Finally, we drop *physician:drug-class* pairings that are almost always zero. For instance, the psychiatrists in the data set will hardly ever prescribe statins. Likewise, cardiologists do not generally prescribe antidepressants. We drop all physician drug-class pairings where the physician prescribed fewer than 48 scripts of all of the drugs in a drug class over the 48-month time period. A physician would have to average fewer than a single script per month of an entire drug class to fall under this threshold. This acts as a participation criterion and eliminates pairings such as cardiologist-antidepressants.

For every physician:drug-class pairing that remains, every physician-drug-month observation is included, even if the physician never prescribes that drug. For instance, if a physician regularly prescribes Effexor but no other antidepressant, that physician would still have observations (with prescriptions values of zero) corresponding to every other antidepressant on the market in each month. These zeroes represent the possibility that the physician could have (but chose not to) prescribe these drugs for patients in need of an antidepressant. The appendix contains many specifications that exclude these rare doctor-drug pairings or otherwise handle the many zeroes in different ways; the results are robust to all of these variations.

3.7. Data Summary

Table 3 displays summary statistics of the final physician-drug-month data set. The high rate of zeroes is not surprising considering that physicians who regularly prescribe a drug are still unlikely to prescribe every drug in that class in every month. The averages in Table 3 make sense when we consider how many drugs each average applies to for each physician-month. For instance, a physician that prescribes only drugs from the antidepressant class and averages 0.6 prescriptions per physician-drug-month

across both generic and branded drugs would be writing 27 scripts per month, because there are up to 45 products in the antidepressant category in a given month. Physicians that average 0.6 prescriptions per drug-month and also prescribe more than one drug class would be writing even more scripts per month on average.

4. Results

4.1. Statistical Method

We use ordinary least squares to implement a difference-in-differences regression model. We choose a linear model in part for ease of interpretability, but also because it can more easily handle the many physician-drug-month observations with zero prescriptions. Nonlinear models will struggle with the variance matrix of these data, which can be highly singular due to the many zeroes.

The dependent variable (DV) of interest is the number of scripts of drug j prescribed by physician i in month t . The independent variables include an indicator for whether the sunshine law is in place for each physician-drug-month observation, which depends on state s and month t . In some specifications, we instead use an indicator for whether the sunshine law has been signed and announced. The regression model is the following, with the dependent variable R_{ijt} representing prescriptions:

$$R_{ijt} = \beta_0 + \beta_1 * \text{sunshine}_{st} + \beta_2 * \text{branded}_{jt} + \beta_3 * (\text{branded}_{jt} * \text{sunshine}_{st}) + X_{ijt} + \varepsilon_{ijt} \quad (1)$$

The sunshine_{st} variable takes a value of 1 if a sunshine law was in effect (or in alternative specifications, has been signed) for that physician-month. The branded_{jt} variable represents whether a drug is branded with no generic available for a given drug-month. As previously discussed, this closely proxies for whether a drug is marketed in a given month.⁸ The $\text{branded}_{jt} * \text{sunshine}_{st}$ interaction separates the effect of the sunshine law on branded drugs from its effect on nonbranded drugs. The coefficient for sunshine_{st} (β_1) measures the latter, and the coefficient for $\text{branded}_{jt} * \text{sunshine}_{st}$ (β_3)

Table 3. Summary Statistics

	Total observations, prelaw (physician-drug-month)	July 2007–June 2009 (prelaw) ^a	Total observations, postlaw (physician-drug-month)	July 2009–July 2011 (postlaw) ^a
Branded Rx, Massachusetts	1,712,756 (90% zeroes)	0.58 (3.43)	1,774,890 (89% zeroes)	0.55 (2.91)
Branded Rx, Control States	1,605,378 (91% zeroes)	0.50 (3.29)	1,651,532 (90% zeroes)	0.54 (3.28)
Generic Rx, Massachusetts	3,103,928 (88% zeroes)	0.74 (4.21)	2,243,127 (82% zeroes)	1.18 (5.31)
Generic Rx, Control States	2,903,544 (91% zeroes)	0.42 (3.11)	2,018,655 (86% zeroes)	0.74 (4.14)

^aThis column reports the mean (standard deviation) of prescriptions per physician-drug-month, for the listed subset of drugs and physicians.

measures the additional effect on branded drugs. To properly interpret the net effect of the sunshine law on a branded drug, we will evaluate the linear combination of the coefficients for sunshine_{st} and $\text{branded}_{jt} \cdot \text{sunshine}_{st}$.

The remaining variables in the vector X_{ijt} are a set of controls including various fixed effects. Our base specification only includes physician fixed effects, and we add a series of fixed effects for drugs, months, and various interactions of physicians, drugs, and months. State and AMC fixed effects are never included; because the sample is limited to physicians that never switch affiliations, physician is collinear to state and AMC. Because including both fixed effects and a lagged dependent variable can complicate identification (Angrist and Pischke 2009), the specifications use fixed effects but no lag.

Standard errors are clustered at the AMC level. Although the disclosure policy and thus treatment is at the state level, we only have five states in the data and clustering over so few states could lead to bias (Cameron and Miller 2015). Instead, we choose to cluster at the AMC level, which can also partially account for institution-level correlations in prescriptions (e.g., philosophical practices, administrative influences, patient demographics, etc.). All results are robust to clustering at the physician or state level, as shown in the appendix (Table A.1).

4.2. Regression Results

Table 4 shows the results of our statistical estimation as we introduce more fixed effects. The baseline regression only uses physician fixed effects; this only controls for underlying physician heterogeneity in prescribing patterns, which previous research shows can be vast (Berndt et al. 2015). To ease interpretation, we report the relevant coefficients and standard errors for the linear combination of β_2 and β_3 ; as noted, this linear combination represents the net effect of the sunshine law on branded, marketed drugs.

With physician fixed effects only (model 1), the coefficient on the linear combination is negative but has a p -value of only 0.14. However, this specification does not control for drug or time trends, and therefore generates considerable noise. As fixed effects controlling for these factors are added, our estimates of the policy effects on marketed drugs become more precise. Models (5) and (6), which use physician*drug and drug*month fixed effects, respectively, are highly conservative because they control for underlying physician prescribing propensity for all drugs (model 5) or for time trends at the individual drug level for all drugs (model 6). All of the models that include any form of drug or month fixed effects (in addition to physician fixed effects) suggest that branded drug prescriptions are statistically significantly lower due to disclosure. In addition, all but one of the models

Table 4. Disclosure Effects, OLS, Physician-Drug-Month

	(1)	(2)	(3)	(4)	(5)	(6)
N (physician-drug-month)	17,013,810	17,013,810	17,013,810	17,013,810	17,013,810	17,013,810
Adjusted R^2	0.060	0.060	0.186	0.186	0.874	0.189
Main Effects						
1. Sunshine Implementation	0.474*** (0.090)	0.284*** (0.097)	0.320*** (0.073)	0.255*** (0.085)	0.041 (0.037)	0.292*** (0.104)
2. Branded	−0.116 (0.078)	−0.129 (0.079)				
Interactions						
3. Sunshine*Branded	−0.584*** (0.159)	−0.571*** (0.159)	−0.481*** (0.143)	−0.483*** (0.143)	−0.181*** (0.025)	−0.562*** (0.182)
Controls						
Month FE	No	Yes	No	Yes	Yes	No
Drug FE	No	No	Yes	Yes	No	No
Physician FE	Yes	Yes	Yes	Yes	No	Yes
Drug*Month FE	No	No	No	No	No	Yes
Physician*Drug FE	No	No	No	No	Yes	No
Linear combinations of coefficients						
a.) 1 + 3	−0.110 (0.072)	−0.288*** (0.069)	−0.161** (0.072)	−0.227*** (0.065)	−0.140*** (0.042)	−0.271*** (0.083)

Notes. The *Branded* indicator is dropped for being collinear whenever drug FE are included. Linear combination (a) represents the net effect of *sunshine-implementation* on *branded drugs*. For comparison, the average branded Rx in MA, postpolicy, was 0.547 in all specifications. We can use this to estimate extrapolated effect sizes; for instance, for Model (1), we calculate $0.110 \div (0.110 + 0.547)$. See Figure A.2 for details of a similar approach. SEs are robust and clustered by AMC.

** $p < 0.05$; *** $p < 0.01$.

(model 5) suggest that generic drug prescriptions increased due to disclosure; these specifications suggest a substitution story between branded and generic drugs.

To be conservative, we focus on model 5. This specification has the smallest estimated net effect on branded drugs. It indicates that the sunshine law decreased branded prescriptions by on average 0.140 scripts per physician-drug-month for every branded drug the physician prescribes. This extrapolates to approximately 248,625 fewer prescriptions of branded drugs for just the Massachusetts physicians, drugs, and months in this data set. This model shows no statistically significant effect of the policy on generic drugs; therefore, the net effect implies that physicians prescribed approximately 6.4% fewer drugs in total. The other models show an increase in generic prescriptions (i.e., a substitution effect), so it is possible that this more conservative model underestimates substitution between branded and generic drugs. The detailed math behind these estimates can be found in the appendix (Figure A.2).

Note that the data include prescription refills, which constitute a significant majority of total prescriptions filled. Patients already on medication that works for them will continue to stay on that medication regardless of disclosure, especially for the chronic conditions treated by the drugs in these data, for example, hypertension, depression, diabetes, high cholesterol, etc. Thus, refills are not likely to be affected, and instead, disclosure will primarily affect first-time prescriptions, leading to a gradual change (i.e., a change in slope) in branded prescriptions as new prescriptions are mixed into the data over time. Thus, the results likely underestimate effects on new prescriptions.

4.3. Pretrends and Timing of Effects

The largest confound for difference-in-differences estimation is that the treatment effect picks up trends that already existed before treatment, or that emerged after treatment due to an unobserved shock. To demonstrate that effects occurred precisely around the time of treatment, it is important to show the trends in treatment effects over time, including quasi-treatment effects in the time periods leading up to treatment.

To do this, we implement a model that separately estimates difference-in-differences coefficients for all 48 months, with the omitted month (and therefore the point of comparison) being June 2008, just over one month before the policy was passed and signed into law on August 10, 2008. The regression equation is:

$$R_{ijt} = \sum_{x=1}^{48} \beta_x \text{MassachusettsDoc}_i * \text{branded}_{jt} * I_{t=x} + \lambda_t + \delta_{ij} + \gamma_{jt} + v_{it} + \varepsilon_{ijt}. \quad (2)$$

In this equation, R_{ijt} and branded_{jt} are the same as before, $\text{MassachusettsDoc}_i$ is an indicator for whether

the physician practices in Massachusetts, and I represents an indicator for whether the month is x , with x ranging from 1 to 48 (representing July 2007 to July 2011). The omitted category is $x = 11$, corresponding to June 2008. The remaining variables represent fixed effects and the error term. In Figure 1, we plot the β coefficients for Massachusetts doctors and branded drugs by month.

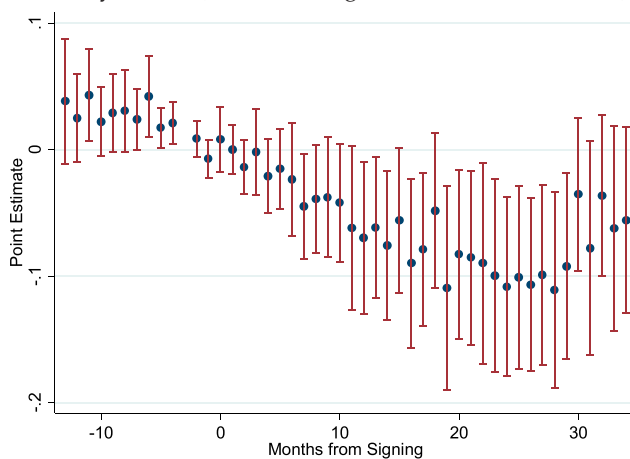
Figure 1 demonstrates that there is no significant trend between groups prior to the omitted month, but a clearly decreasing trend afterward. It does appear that branded prescriptions may slightly drop just before the month of signing. Given the likelihood that the law was discussed among the medical community prior to being signed, this is not too surprising. Overall, the figure supports the fixed differences assumption of the identification strategy. Branded prescriptions for Massachusetts physicians trends down primarily only after the law is signed; this is consistent with the previously observed effects in Table 4.

We further test the timing of the effect by using the signing date of August 2008 as the treatment shock, rather than the implementation of disclosure. Results are shown in Table 5 and are consistent with Figure 1, with less precise and smaller, but still significant, treatment effects. The data are consistent with a story where physician behavior changed somewhat after the policy was signed into law, but where the majority of the change happened once payments data were subject to be disclosed.

4.4. Robustness Checks

To address the possibility of false positives due to serial correlation in the error term (Bertrand et al. 2004), we use two strategies. First, we cluster standard errors at different levels. Table A.1 in the appendix reports these alternate standard error specifications.

Figure 1. (Color online) Pretreatment and Posttreatment Effects by Month (Branded Drugs, Massachusetts Doctors)



Notes: Omitted month is June 2008, five weeks prior to the law being signed. Law implementation (the month it takes effect) occurs at month 11.

Table 5. Disclosure Effects, Law Signed

	(7) Rx
DV	
N (physician-drug-month)	17,013,810
Adjusted R^2	0.874
1. Sunshine Signed/Announced	0.033 (0.024)
2. Sunshine*Branded	−0.140*** (0.022)
Controls	
Month FE	Yes
Physician*Drug FE	Yes
Linear combinations of coefficients	
a.) 1 + 2	−0.107*** (0.029)

Notes. SEs are robust and clustered by AMC. Linear combination (a) represents the net effect of *sunshine-announced* on *branded drugs*. For comparison, the average branded Rx in MA, postsigning, was 0.552. *** $p < 0.01$.

Results are robust to clustering by physician, clustering by state, and to block-bootstrapping.

Placebo tests that randomly assign the time of treatment, and/or the treated observations, are also frequently used to assess robustness of difference-in-differences models, particularly due to serial correlation and false positives (Bertrand et al. 2004). We run a placebo test by excluding all Massachusetts doctors and randomly assigning half of the counterfactual doctors to the treatment group, that is, a disclosure policy in July 2009. We then run the preferred specification (model 5 of Table 4) on this placebo treatment group. We repeat this 100 times and plot the estimated policy effect on branded drugs in Figure 2. Statistical significance was achieved in only 3 out of 100 trials.

Tables A.1 through A.16 in the appendix include additional robustness checks. Results are robust to dropping any counterfactual state, to including all drugs previously dropped per the steps in the data exclusions section, and to controlling for additional data on AMC-specific policies on pharmaceutical marketing. Results are also similar when excluding all generic drugs, when collapsing the data down to the doctor-month level (thus ensuring the results were not driven by the many zeroes in the drug-level data), and when dropping rare doctor-drug pairs. Table A.10 demonstrates that effects are stronger for highly prescribed drugs, which (due to drugs such as Lipitor, Nexium, and Diovan) are concentrated more in the nonpsych classes. Results are also robust to using $\log(\text{prescriptions})$. Finally, results are also robust to examining only the drugs that are present for the full 48 months of our data. This demonstrates that the results are not being influenced by drugs coming in

or out of the sample,⁹ and the unequal prepolicy and postpolicy sample sizes that these observations induce (see Table A.18).

4.5. Mechanisms

Without prepolicy payments data, we cannot directly examine whether disclosure impacts prescriptions by changing whether or how much a physician accepts. However, we can take advantage of drug and physician heterogeneity to examine whether disclosure is affecting the drugs and physicians most likely to be targeted by pharmaceutical meals and consulting payments. This can test whether our results are consistent with a mechanism whereby disclosure influences willingness to accept payments, leading to changes in prescriptions.

First, we incorporate estimated drug-level national sales force size data from the second quarter (April to June) of 2009. Some drugs that are detailed nevertheless do not have national dedicated sales forces, indicating they are not as heavily marketed, whereas others will have national sales forces of differing sizes, depending on how heavily marketed they are. Model 8 in Table 6 runs the preferred specification from before, but using an indicator to separately estimate effects for drugs with and without a national sales force as of the second quarter of 2009. The results suggest that drugs with sales forces are much more significantly affected by disclosure. Model 9 replaces the binary indicator for sales force with the log of the actual sales force size, and it shows that the larger the sales force, the more heavily a drug was affected by disclosure. This implies that the more marketed a drug was (nationally) in the months leading up to disclosure, the stronger the subsequent estimated effect of disclosure on that drug.

Figure 2. (Color online) Placebo Test, Branded Drugs, 100 Trials

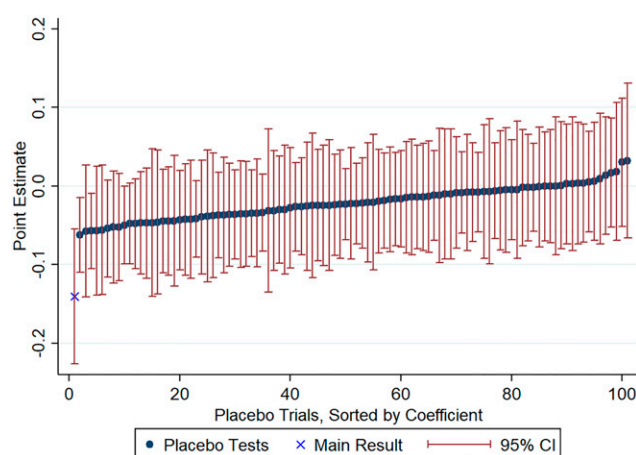


Table 6. Drug Sales Force Size

DV	(8)	(9)
	Rx	Rx
N (physician-drug-month)	17,013,810	17,013,810
Adjusted R^2	0.874	0.874
Main Policy Effects		
1. Sunshine Implementation	0.041 (0.037)	0.041 (0.037)
Interactions		
2. Sunshine*Branded	−0.115*** (0.030)	−0.116*** (0.029)
3. Sunshine*Any Sales Force	−0.132* (0.071)	
4. Sunshine*Ln(Sales Force)		−0.017* (0.009)
Controls		
Month FE	Yes	Yes
Physician*Drug FE	Yes	Yes
Linear combinations of coefficients		
a.) 1 + 2 (Branded)	−0.074* (0.043)	−0.074* (0.042)
b.) 1 + 2 + 3 (Branded + Sales Force)	−0.205*** (0.064)	

Notes. SEs are robust and clustered by AMC. Linear combination (a) represents the net effect of *sunshine* on *branded drugs* with no sales force. Linear combination (b) represents the net effect of *sunshine* on *branded drugs* with a sales force. For comparison, the average branded Rx in MA, postpolicy, was 0.157 for those with no sales force, and 1.228 for those with a sales force.

* $p < 0.10$; *** $p < 0.01$.

We next examine treatment effects across doctor types based on their pretreatment prescribing volume. Pharmaceutical companies purchase the same IMS data used in this study, and they focus their marketing predominantly on high-prescribing physicians, and especially those who prescribe many marketed drugs (Fugh-Berman and Ahari 2007). High prescribing volume prior to the sunshine law is therefore an exogenous indicator that is correlated with receiving payments in the pretreatment period. We divide our physicians into halves based on pretreatment prescribing. Because as many as 70% of physicians may accept at least one meal or gift from salespeople per year (Campbell et al. 2010), the top half is likely to encompass many physicians with prepolicy payments, whereas the bottom half likely contains many fewer such physicians. Models 10 and 11 in Table 7 separately estimate the effects of disclosure when splitting physicians by total prepolicy prescriptions, and by total prepolicy branded prescriptions, respectively. Model 12 also splits by total prepolicy prescriptions, but separately for each drug class. The linear combination terms in all three specifications imply that the effect of disclosure on branded drugs occurs only on physicians above the median in prepolicy prescribing volume. Physicians in the bottom half do not change behavior, and this null

effect is fairly precisely estimated. Therefore, disclosure affects precisely the drugs and physicians that were most likely to be targeted by pharmaceutical sales people.

The extensive experimental literature on disclosure suggests that patients are unlikely to drive responses to disclosure. To test whether patients played a key role in the observed effects, we examine whether the results are robust to excluding data from after November 2010, the month the first batch of data were released (Sullivan 2011) and the only time data were released during this time period. We can also examine whether results are similar when including data after November 2010, but excluding physicians that had payments disclosed in the November 2010 data. Both regressions are displayed in Table 8. In both cases, results are largely similar to before. This suggests that patient response to the disclosed payments is not necessary for disclosure to impact prescriptions. Similarly, these results also cannot be explained by others (e.g., media, colleagues, hospital administrators, etc.) responding to the disclosed payments.

Instead, our evidence is consistent with a physician image mechanism, whereby physicians want to avoid being perceived (by patients, colleagues, hospital executives, etc.) as biased or unethical. They may respond to disclosure by opting out of or reducing

Table 7. Splitting Doctors by Prepolicy Prescriptions

	(10)	(11)	(12)
DV	Rx	Rx	Rx
Prepolicy halves split by:	Total Rx	Total branded Rx	Total class Rx
N (physician-drug-month)	17,013,810	17,013,810	17,013,810
Adjusted R^2	0.874	0.874	0.874
Policy*Half			
1. Sunshine Implementation*Top	0.027 (0.041)	0.027 (0.042)	0.041 (0.039)
2. Sunshine*Bottom	0.086* (0.043)	0.074** (0.035)	0.150*** (0.049)
Policy*Half*Branded			
3. Sunshine*Top*Branded	−0.210*** (0.033)	−0.213*** (0.031)	−0.213*** (0.044)
4. Sunshine*Bottom*Branded	−0.072 (0.054)	−0.078** (0.035)	−0.228*** (0.027)
Controls			
Month FE	Yes	Yes	Yes
Physician*Drug FE	Yes	Yes	Yes
Linear combinations of coefficients			
a.) 1 + 3 (Top)	−0.183*** (0.048)	−0.186*** (0.046)	−0.172*** (0.054)
b.) 2 + 4 (Bottom)	0.014 (0.037)	−0.004 (0.036)	−0.078 (0.048)

Notes. SEs are robust and clustered by AMC. Linear combination (a) represents the net effect of *sunshine* on *branded drugs* for the top half. Linear combination (b) represents the net effect of *sunshine* on *branded drugs* for the bottom half. For comparison, the average branded Rx in MA, postpolicy, for the top half was 0.648, 0.684, and 0.806 respectively, for the three models; for the bottom half, it was 0.176, 0.127, and 0.355, respectively.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

industry payments, leading to decreases in branded prescriptions (Larkin et al. 2017). This is consistent with our results being driven primarily by the drugs and physicians most targeted by salespeople. Alternatively, it is possible that disclosure reduces biases and improves advice regardless of whether it reduced a physician's payments or meals from industry, perhaps because disclosure calls to mind norms of patient trust within medicine (Sah 2019) or because it increases the salience of overprescribing. This alternative mechanism can still be consistent with a physician image mechanism, because it may imply that physicians care about whether their prescriptions will appear biased or affect patient trust. Thus, physician image concerns may reduce prescriptions both through reductions in payments and/or through these other mechanisms.

The opt-out mechanism is also consistent with other literature on physician disclosure. Pham-Kanter et al. (2012) find no effect of disclosure to the state government, perhaps implying that the public nature of disclosure is crucial to how physicians respond. Parker (2017) finds a small, positive effect of disclosure on marketed drugs, perhaps because in their context, physicians had no forewarning their

payments would be disclosed and had no option to opt out of payments in advance.

Physician social image concerns may be financial (e.g., if the number of patients and procedures depends on image) or psychological (e.g., disutility from being perceived as unethical). Because these physicians are full-time attendings that are likely salaried, they are less likely than private practitioners to be motivated by patient or procedure volume; one recent estimate holds that AMC-affiliated doctors make several million dollars less in compensation over their careers due to these differences in compensation (Davis 2016). Thus, it is possible their image concerns are at least partially psychological in nature. Indeed, physicians seem to care about (and display emotional responses to) how their patients and others perceive them ethically, independently from any impact on earnings, as seen in physician responses to other disclosure efforts (e.g., Wen 2013a, b; 2014).

4.6. Alternative Mechanisms

Alternatively, it could be that industry (and not physicians) instigated these changes in industry-physician interactions, leading to the observed results. However, this appears unlikely for several reasons. First, as the

Table 8. Disclosure Effects, Absent Patient-Driven Mechanisms

DV	(13)	(14)
	Rx	Rx
Excluded Data	Rx from November 2010 or later	Physicians with payments in first batch
N (physician-drug-month)	14,358,192	15,965,289
Physicians	5,730	5,420
Adjusted R^2	0.887	0.868
Policies		
1. Sunshine Implementation	0.055 (0.034)	0.040 (0.044)
Interactions		
2. Sunshine*Branded	−0.160*** (0.026)	−0.179*** (0.028)
Controls		
Month FE	Yes	Yes
Physician*Drug FE	Yes	Yes
Linear combinations of coefficients		
a.) 1 + 2	−0.106*** (0.038)	−0.139*** (0.041)

SEs are robust and clustered by AMC. Linear combination (a) represents the net effect of *sunshine* on *branded* drugs. For comparison, the average branded Rx in MA, postpolicy, was 0.580 and 0.516, respectively.

*** $p < 0.01$.

disclosures data illustrate, pharmaceutical firms continued to provide significant numbers of meals and consulting payments to Massachusetts physicians postlaw.¹⁰ Second, despite our efforts, we could find no industry source that ever suggested implementing a state-level change in marketing tactics (such as a state-specific change in emphasis on marketing activities or sales force size) in response to this policy. Instead, news interviews with industry representatives imply that salesperson marketing strategies are not state-specific, and moreover that national-level changes in physician-industry interactions are attributed to changing physician attitudes toward salespeople and not to any state-specific laws (Rockoff 2012). Recent national-level decreases in various marketing activities have also been publicly attributed to loss of exclusivity (i.e., patent expiration) of important drugs (Sullivan 2013), with no mention of disclosure policies.¹¹

Another possible interpretation could be that effects are due to the Massachusetts healthcare bill signed in April 2006 (colloquially known as “Romneycare”) that required nearly all Massachusetts residents to obtain a minimum level of health insurance coverage. However, if this bill affected prescriptions, it would likely be predominantly in the year the bill took effect (in 2006), more than a year before our data set begins and more than three years from the implementation of disclosure. This is because 2006 is when most residents would have signed up for healthcare to avoid the penalty. Moreover, an increase in health insurance coverage could *increase* branded drug prescriptions because increased insurance coverage makes the

expensive, branded drugs more affordable to patients.¹² These factors make it unlikely that this bill would drive the observed decrease in branded drug prescriptions in our data.

A final alternative mechanism could be that these effects were driven by a state-specific health insurance change. In 2009, the insurance company Blue Cross Blue Shield of Massachusetts launched a new payment arrangement, known as the Alternative Quality Contract (AQC). These contracts stipulate fixed payments for the care of a patient over a specified time period, and they connect payments to quality goals and a five-year budget (Chernew et al. 2011). However, AQCs covered only 2,300 physicians, or about 8% of physicians in the state, at the time of disclosure (McPhee 2010), though this approximately doubles by end of 2010 (Chernew et al. 2011). Most importantly, these changes are unlikely to lead to changes in prescriptions; Afendulis et al. (2014) used a difference-in-differences approach comparing drug prescription usage by Massachusetts physicians belonging to providers that enrolled versus did not enroll in an AQC in 2009. They found that AQCs in Massachusetts did not have any impact on the use of either branded or generic drugs, with a fairly precisely estimated null effect for both branded and generic drugs. The authors provided several reasons why these results make sense and why, if anything, AQCs could raise branded drug prescriptions, because they provide incentives to meet various biometric markers (such as blood pressure or cholesterol ratings) that are best controlled via drugs (ibid).

5. Discussion

5.1. Summary

This paper uses a state-level policy change to evaluate whether mandated disclosure of payments from pharmaceutical companies changed physician prescribing. The results suggest that public disclosure reduced the prescriptions of branded drugs, and some specifications also suggest a corresponding increase in generic prescriptions (i.e., a substitution effect). Importantly, these effects were driven predominantly by the drugs and physicians most likely to have been influenced by pharmaceutical meals and payments. In addition, the results cannot be explained solely by patients (or others) responding to the information that was disclosed, because effects are detectable before the first data were released, and because many affected physicians did not disclose any payments.

Taken together, our results are most consistent with a story whereby disclosure invokes image concerns in physicians—a mechanism that has been demonstrated in the laboratory and in survey-based literatures on disclosure. Physicians are trained in the spirit of the Hippocratic Oath, and disclosure may interact with this norm and encourage physicians to reduce or abstain from meals and payments to avoid appearing biased, potentially leading to decreases in branded drug prescriptions (Sah and Loewenstein 2014). Alternatively, physician concern over appearing biased could lead to fewer prescriptions of heavily marketed drugs, regardless of changes to interactions with industry (Sah 2019).

These results help overcome contradictory findings in the current empirical literature. The implicated physician-image mechanism was not likely in two of the prior studies on disclosure, because disclosure was either not to the public (Pham-Kanter et al. 2012) or was not anticipated by physicians (Parker 2017). Unlike Guo et al. (2019), our results also implicate a mechanism more closely tied to influence by marketing activities and payments, perhaps because propensity to overprescribe in general (the mechanism identified by Guo et al. 2019) may differ between our physician samples. Finally, relative to past studies, we are the first to analyze predisclosure drug and doctor heterogeneity to better isolate the mechanisms, and we also use the most comprehensive prescriptions data and the most drug classes.

Altogether, our results provide an important counter to those claiming that disclosure is ineffective, either due to lack of patient awareness of disclosure (Lenzer 2016) or due to the pitfalls of disclosure (Barbour et al. 2012). Despite these concerns, our results demonstrate that when conflicting incentives are voluntary or optional, disclosure can still be effective by causing agents to change behavior to avoid appearing biased.

5.2. Limitations

The study has a number of limitations. Most importantly, we do not observe prepolicy payment data for the treatment group, and we do not observe any payment data for the control group. We therefore used prepolicy prescribing levels and prepolicy national sales force sizes as proxies for which doctors and drugs were most likely targeted for payments; the two variables are demonstrated to be highly correlated in the literature, but more complete payments data would isolate the mechanism more directly.

Like any paper using difference-in-differences techniques, we cannot definitively show the treatment estimate is caused by the adoption of policies. The effect of other shocks contemporaneous to the new policy will show up as part of our treatment estimate. Although we attempted to rule out major alternative explanations, we cannot definitively show causality.

Finally, we cannot say for certain how our treatment estimates would hold for nonacademic doctors. Academic physicians are more involved in movements such as promoting evidence-based medicine or academic detailing (e.g., Fischer and Avorn 2012), and this might mean that the effect of disclosure will be even higher for the ~85% of U.S. physicians in private practice. However, one might also argue that the social image mechanisms suggested by our results would not be as keenly felt for nonacademic physicians, which would mean that physicians in private practice would not be as influenced.

5.3. Managerial Implications

These results carry important managerial implications in healthcare. For healthcare managers and officials concerned with the effects of pharmaceutical marketing on prescription drug costs, increasing the coverage of disclosure or making disclosed payments more salient (e.g., by implementing hospital-wide communications or campaigns) may be an effective method for changing physician behavior. Other physician conflicts of interest may also benefit from disclosure. For instance, the “Total Transparency Manifesto” and the “Who’s My Doctor?” campaigns both advocate for physicians to disclose all sources of potentially conflicting incentives, including incentives for ordering additional tests or procedures (Wen 2013a, b; Sifferlin 2014). Approximately 70% of surveyed physicians believed that clinicians are more likely to perform unnecessary procedures when they profit from them (Lyu et al. 2017); disclosure of these payments may be worth exploring, especially as alternative pay structures continue to be introduced into the field, thus making fee-for-procedure structures more optional.

These results may also carry important implications for how managers and officials manage conflicts

of interest even in nonhealthcare settings. The principal-agent problems inherent in drug prescribing, where an informed expert makes important decisions for an uninformed principal, are found in many other industry settings such as retirement planning, consumer insurance, mortgage origination, and legal advice, to name a few. However, within medicine, there are norms (such as the Hippocratic Oath) that place great importance on earning a patient's trust (Sah 2019); because our results suggest agents must care about appearing unbiased in order for disclosure to work, it remains for future research to test whether disclosure is effective in settings where such norms are not as heavily emphasized. Nevertheless, the results in this paper suggest that disclosure is at least worth exploring further in these contexts, despite the literature on the pitfalls of disclosure.

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Endnotes

¹ For instance, the current federal disclosure policy does not require reporting of free meals or payments to nonphysicians, but according to former salespeople, office staff are often targeted due to their ability to influence physicians (Fugh-Berman and Ahari 2007). Details of the policy are available at: <https://www.cms.gov/OpenPayments/Downloads/open-payments-general-faq.pdf>.

² This would potentially run counter to the quote from the Restaurant and Business Alliance. However, it could be that pharmaceutical companies decreased meals or payments in response to a decrease in prescriptions.

³ Minnesota, Washington D.C., and Vermont also implemented disclosure policies (requiring disclosure to the government) prior to Massachusetts, but their disclosures information was not collated into a publicly accessible electronic data set and often proved difficult even for researchers to access (Ross et al. 2007, Chimonas et al. 2010).

⁴ Our results may differ because of the different types of physicians in our samples. Our sample consists only of attending physicians at academic medical centers (AMCs); these are highly successful physicians working full time at renowned hospitals such as Mass General or Mount Sinai, and they may be less prone to general overprescribing (e.g., of opioids, antihyperactives, and the like) than the private practitioners in Guo et al. (2019), leading to differences in effects and mechanism. However, this is only speculative because neither study included physicians of both types.

⁵ These physicians may have been erroneously categorized as attendings (instead of admitting) by IMS during their quarterly surveys; some had multiple affiliations to different locations in the same quarter. Excluding these physicians also rules out the possibility that any physicians switched affiliations in anticipation of the policy. These exclusions affected a small fraction of physicians that met our other criteria for inclusion.

⁶ For instance, during the entire timespan of our data, Prozac had a generic alternative (fluoxetine hcl) available, and 99% of all prescriptions for Prozac/fluoxetine were for the generic version.

⁷ In contrast, insurance companies cannot legally mandate interchange of nonbiologically equivalent drugs. Also, insurance policies on reimbursement denial for similar but nonbiologically equivalent drugs do not vary by state (Brown 2016).

⁸ Drugs that are branded but which have a generic available in that month take on values of 0 because evidence suggests detailing stops once a generic is available, due to mandated interchange to generics by insurance companies, states, and other regulatory agencies.

⁹ Recall that these drugs are added mid-sample either because that is when the drug is first introduced to the market, or because the drug was included in one of our two data purchases but not both. (The latter occurred only if they were not frequently prescribed by our physicians in those years, according to IMS Health.)

¹⁰ The sunshine law required pharmaceutical firms to track and report these payments, which entails a mostly fixed cost of establishing a tracking system; once this fixed cost is paid, there is little additional cost of tracking to increasing the number of meals or payments made. It should also be noted that most pharmaceutical companies likely already had tracking systems in place in order to evaluate the effectiveness of their marketing practices.

¹¹ For our results to be industry driven, it would imply that industry changed marketing prior to the first release of payments data, given our results in Table 8. Because industry is unlikely to forego a year's worth of profit from marketing, this would in turn suggest that industry expected disclosure would change the effectiveness of their marketing prior to the data being released, implying expectations of a doctor-driven response to disclosure.

¹² One concern could be if insurance companies reduced coverage of prescription drugs in response to Romneycare. However, this bill was motivated in part by the fact that many residents were scheduled to lose Medicaid coverage due to an expiring federal government waiver, and more than half of new enrollees under this bill were through Medicaid. As documented on the federal government website, during this time span Massachusetts did not change Medicaid coverage of the drugs in this data set, except for a small group of anxiolytics and benzodiazepines, and our results are robust to excluding these. In addition, counterfactual states including California and Illinois implemented similar changes to their Medicaid coverage at the same time as Massachusetts, suggesting these were not motivated by disclosure (<https://www.medicaid.gov/medicaid/prescription-drugs/excluded-drug-coverage/index.html>).

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